पेटेंट कार्यालय
शासकीय जर्नल

OFFICIAL JOURNAL
OF
THE PATENT OFFICE

निर्गमन सं. 34/2026	शुक्रवार	दिनांक: 21/08/2026
ISSUE NO. 34/2026	FRIDAY	DATE: 21/08/2026

पेटेंट कार्यालय का एक प्रकाशन
PUBLICATION OF THE PATENT OFFICE

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :30/06/2026

(21) Application No.202621080930 A

(43) Publication Date : 21/08/2026

(54) Title of the invention : DIGITAL TWIN BASED SYSTEM AND METHOD FOR PERSONALIZED LUNG CANCER TREATMENT IN METAVERSE ENVIRONMENTS

(51) International classification	:G16H 30/40, G16H 50/50, G16H 30/20, G16H 50/20, G06T 19/00
(31) Priority Document No	:NA
(32) Priority Date	:NA
(33) Name of priority country	:NA
(86) International Application No Filing Date	: :01/01/1900
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

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(57) Abstract :
ABSTRACT DIGITAL TWIN BASED SYSTEM AND METHOD FOR PERSONALIZED LUNG CANCER TREATMENT IN METAVERSE ENVIRONMENTS
The present invention provides a digital twin based system (100) and method for personalized lung cancer treatment simulation in a metaverse environment. The system (100) comprises a data acquisition and preprocessing module (110) for processing DICOM CT images with Hounsfield Unit conversion and isotropic resampling, a segmentation extraction module (120) for extracting gross tumor volume masks from DICOM SEG objects, a three dimensional mesh generation module (130) employing Marching Cubes algorithm and Laplacian smoothing, a radiomic feature extraction module (140) computing over one hundred quantitative features, a machine learning inference module (150) using Random Forest classifiers for risk stratification and survival prediction, a treatment simulation engine (160) implementing Gompertz differential growth equation with treatment specific kill rates for radiotherapy, chemotherapy, immunotherapy, and surgery, a toxicity modeling module (170) with cyclic Gaussian pulse and linear accumulation functions, and a metaverse visualization interface (180) providing immersive three dimensional rendering. [to be published with fig.1]

No. of Pages : 30 No. of Claims : 10